

Code-Layout, Readability and Reusability

Date:	15 March 2026
Team ID:	(xxx)
Project Name:	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks:	5

Code-Layout, Readability and Reusability

The **Human Development Index (HDI) Prediction System** is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript** in **Visual Studio Code (VS Code)**. The project follows good coding practices by organizing files into separate folders, using meaningful variable names, writing reusable functions, and adding comments where required. This improves readability, maintainability, and future scalability.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code.	Yes	Improves readability and follows Python coding standards (PEP 8).
2	Proper File Structure	Files and folders are logically organized.	Yes	Separate folders for templates, static, dataset, and models.
3	Meaningful Variable Names	Variables clearly represent their purpose.	Yes	Variables such as life_expectancy, gni_per_capita, and prediction_result are descriptive.
4	Function / Method Names	Functions are named according to their functionality.	Yes	Functions like load_model(), preprocess_data(), and predict_hdi() improve code clarity.

S.No	Code Quality Parameter	Description	Followed	Remarks
5	Code Comments	Comments explain important logic and processing steps.	Yes	Comments are added for data loading, preprocessing, prediction, and Flask routes.
6	Modular Design	Code is divided into reusable modules and functions.	Yes	Model training, prediction, and web application logic are organized into separate modules.
7	No Redundant Code	Duplicate or unused code has been removed.	Yes	Common logic is reused through functions, avoiding repetition.
8	Error Handling	Input validation and exceptions are handled properly.	Yes	The application checks invalid inputs and handles runtime errors gracefully.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python, Pandas	Used during dataset preparation and prediction	High
2	Machine Learning Model	Scikit-learn	Used for training and prediction	High
3	Flask Application	Flask (Python)	Handles user requests and prediction output	High
4	HTML Form	HTML5, Bootstrap	Collects user input in the web interface	Medium
5	CSS Stylesheet	CSS3	Used across all web pages for consistent design	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project folders and files are well organized, making the application easy to maintain.
Readability	5	Meaningful variable names, clear function names, and proper formatting improve readability.
Reusability	5	Modular functions and reusable components simplify maintenance and future enhancements.
Documentation / Comments	4	Sufficient comments explain the main logic, preprocessing steps, and prediction workflow.
Overall Score	4.8 / 5	The project follows good coding standards and is suitable for development and maintenance in VS Code.

Team Members and Responsibilities

Team Member	Role	Responsibility
G Naveen	Team Lead	Project planning, GitHub repository management, and model integration.
Chandana Busireddy	Member	Dataset collection, preprocessing, and documentation.
Gandham Joshna	Member	Data visualization and exploratory data analysis (EDA).
Nandineni Gowthami	Member	Flask web application development and testing.
Shaik Talha	Member	Machine learning model training, evaluation, GitHub updates, report preparation, and final presentation.

VS Code Project Structure

HDI-Prediction/

```
|  
|└─ dataset/  
|  └─ hdi_dataset.csv  
|└─ models/  
|  └─ model.pkl  
|└─ templates/  
|  └─ index.html  
|└─ static/  
|  └─ style.css  
|  └─ script.js  
|└─ app.py  
|└─ train_model.py  
|└─ requirements.txt  
|└─ README.md  
└─ .gitignore
```